

September
7th-9th

Robotics Software Engineering 2026



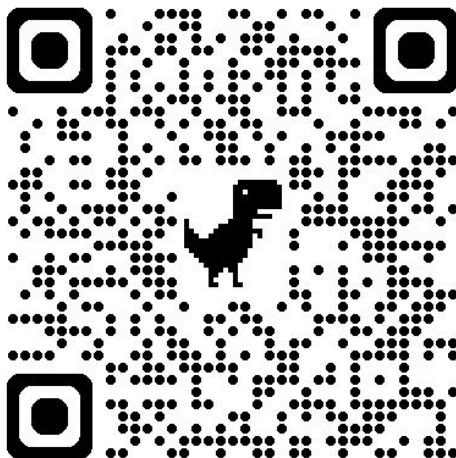
Operationalizing Ethical Rules in Autonomous Robotic Systems

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Reference



R. Casadei, M. De Sanctis, G. Filippone, S. Pettinari, G.L. Scoccia, N. Troquard. **Operationalising Normative Rules in Autonomous Robotic Systems through Context-Oriented Programming.** *International Joint Conference on Artificial Intelligence*. 2026



Part of a bigger project...



Human, societal and environmental
values



Values & Personal Ethics



Human values
Societal and environmental values
Stakeholder roles (proactive, reactive, passive)

SOFT ETHICS

Individual preferences,
context-dependent trade-offs

Impact & Augmentation Analysis



Analyze social and ethical impacts
Human-AI augmentation and replacement
Context- and task-aware considerations

Ethical-Aware Architectures



Allocate responsibilities to system components
Design for transparency, accountability and control
Integrate ethical concerns in system design

Normative Requirements



Express values and norms as requirements
Formalize ethical rules (e.g., with FRET)
Enable analysis, verification and validation

Formal Reasoning & Tooling



Automated analysis of ethical requirements
Tool support (e.g., FRET)
Reason about compliance, trade-offs and conflicts

Runtime Assurance & Enforcement



Monitor system behavior
Enforce ethical rules at runtime
Detect and mitigate misalignments
Support safe and trustworthy operation

Transparency & Ethics Labels



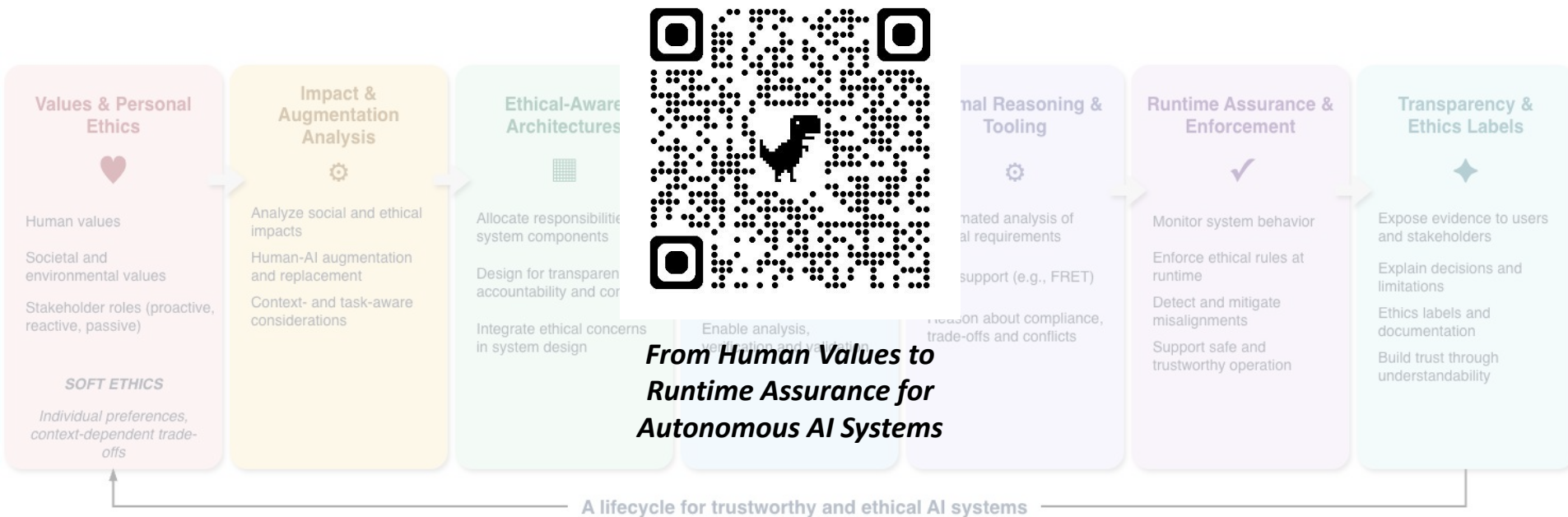
Expose evidence to users and stakeholders
Explain decisions and limitations
Ethics labels and documentation
Build trust through understandability

A lifecycle for trustworthy and ethical AI systems

Part of a bigger project...



Human, societal and environmental
values



Background

- AI and autonomous systems are now widespread
- Concerns about potential harms with their interactions
- Act according to recognized ethical norms, guidelines, principles



Social



Legal



Ethical



Empathetic



Cultural

SLEEC rules have been proposed to elicit social, legal, ethical, empathetic, and cultural norms for autonomous systems

SLEEC rules

G	S
S	I

- Specify behavioral constraints

WHEN *trigger* **THEN** *response*
UNLESS *condition*₁ **IN WHICH CASE** *obligation*₁
...
UNLESS *condition*_n **IN WHICH CASE** *obligation*_n

 default rule

 hedge clause - 1

 hedge clause - n

- Contextualized with system capabilities and domain concerns
- Upon a trigger, a default rule is executed *unless* a contextual condition is triggered, overriding the default response

SLEEC rules



WHEN Curtain open request **THEN** Open curtains

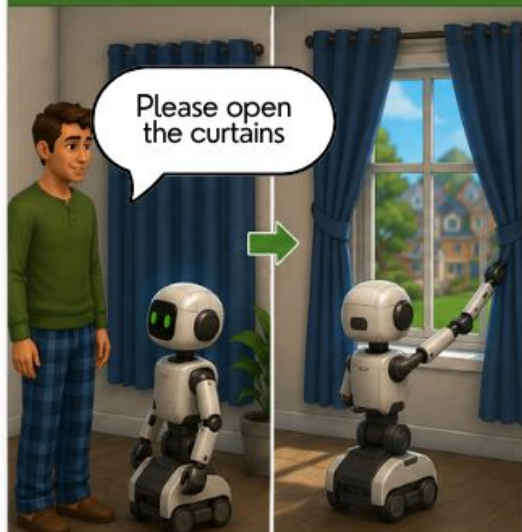
UNLESS User undressed **IN WHICH CASE** Do not open curtains

UNLESS User is distressed **IN WHICH CASE** Open curtains and warn

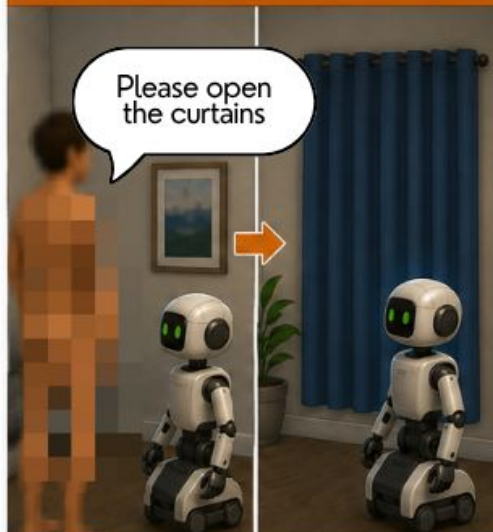
+ Privacy
- Autonomy

+ Mental Health
+ Explainability
- Privacy

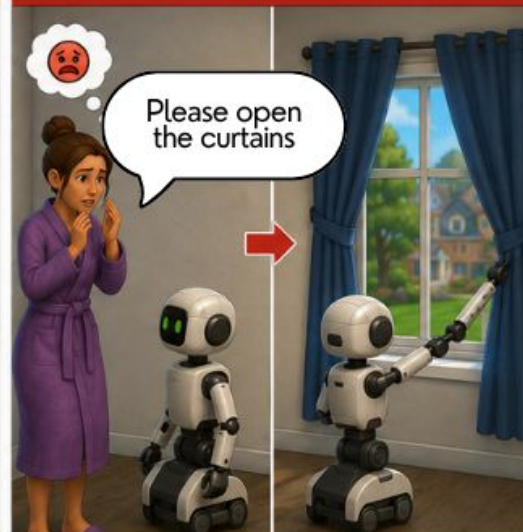
Normal request



User undressed (hedge clause)



User is distressed (hedge clause)

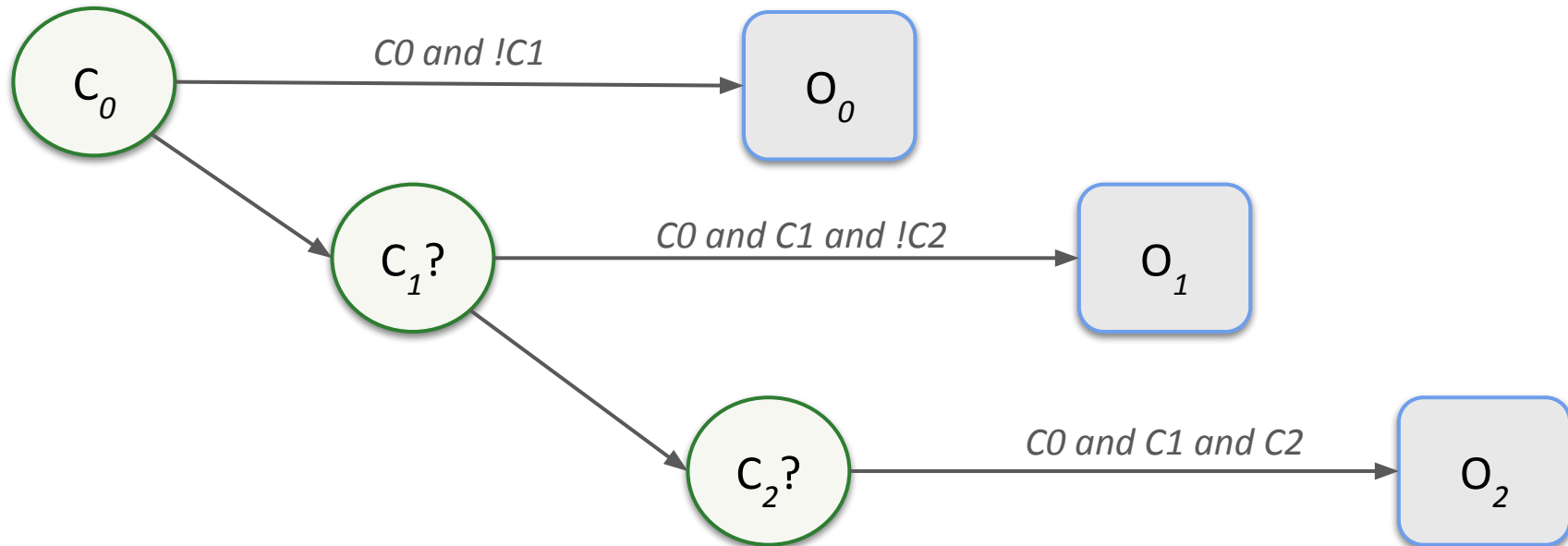


SLEEC rules: operational semantics

WHEN C_0 THEN O_0
UNLESS C_1 IN WHICH CASE O_1
UNLESS C_2 IN WHICH CASE O_2

G	S
S	I

Hedge clauses are checked top-down; the last satisfied one wins



Research gap

State of the art

Approaches for translating high-level principles into SLEEC rules

Focus on early stages: elicitation, formalization, verification

Lack of contributions for *operationalization of SLEEC rules in autonomous systems**

* *operationalization*:

mechanisms and procedures to turn principles into design, implementation and runtime operation



GAP

Key Idea

SLEEC rules ground ethical concerns in specific execution contexts

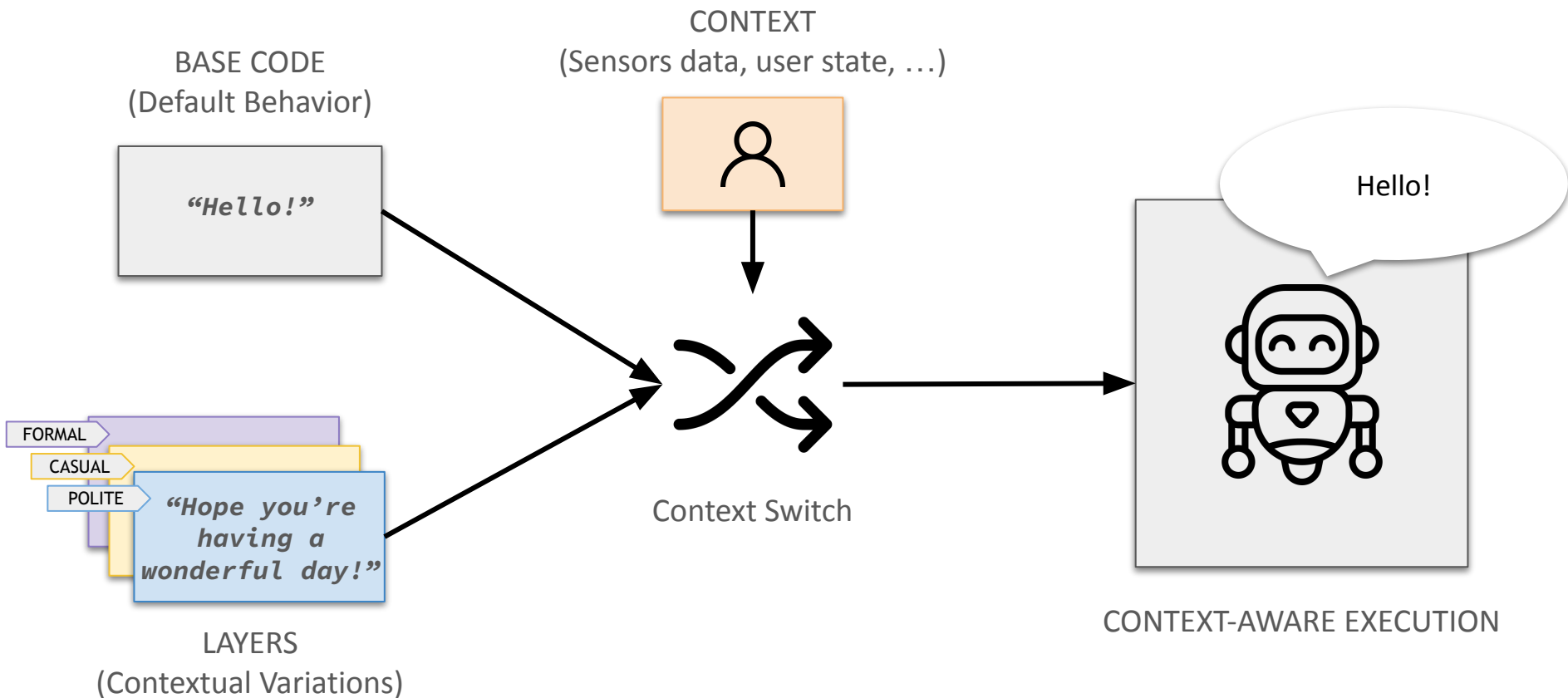


Operationalization requires programming abstractions that explicitly represent context and support runtime adaptation

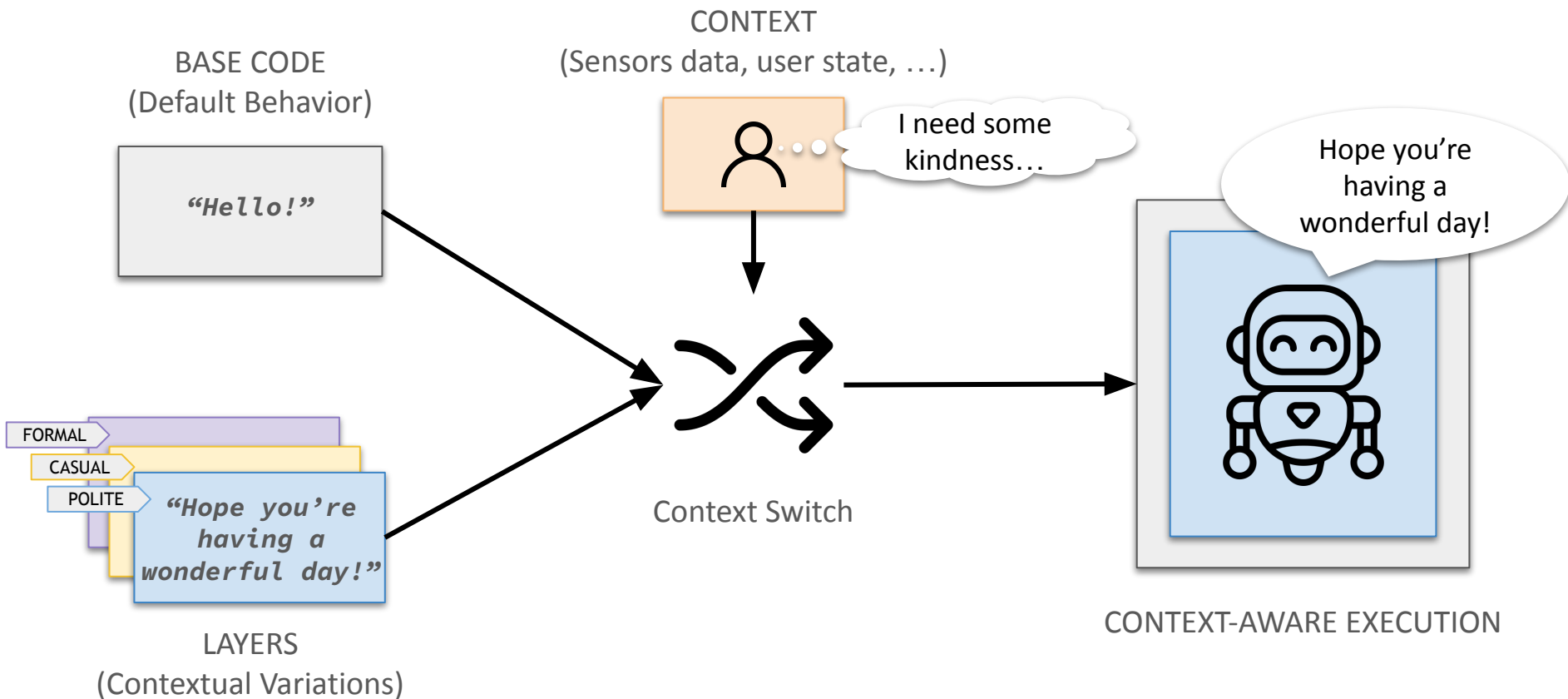


Context-Oriented Programming can be a technological solution enabling context-aware ethical behavior

Context-Oriented Programming (COP)



Context-Oriented Programming (COP)



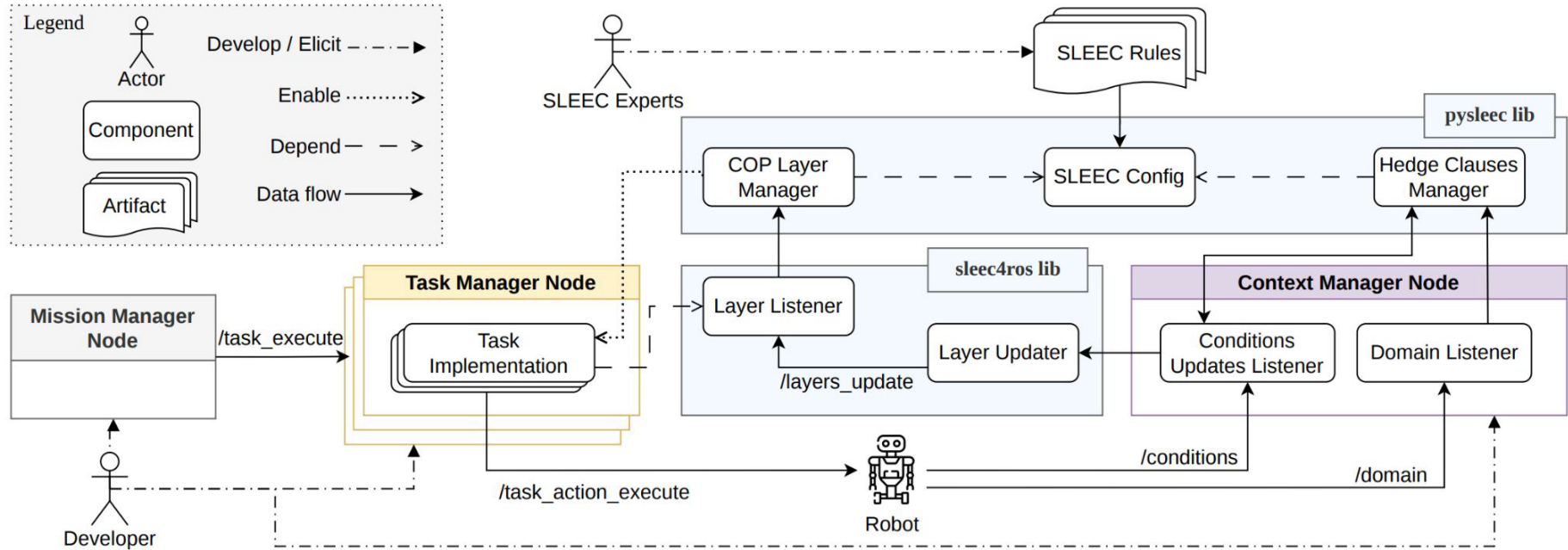
CO-SLEEC approach



Instead of burying norms inside many if-statements, CO-SLEEC modularizes ethical hedge clauses as layers that can be activated by context

SLEEC Elements	COP Concepts
Default rule trigger	Event triggering a method call
Default rule response	Base method implementation
Hedge clause	Layer
Hedge clause condition	Runtime context
Hedge clause obligation	Layer-specific method implementation

CO-SLEEC for ROS architecture



CO-SLEEC for ROS architecture

Reusable Python libraries

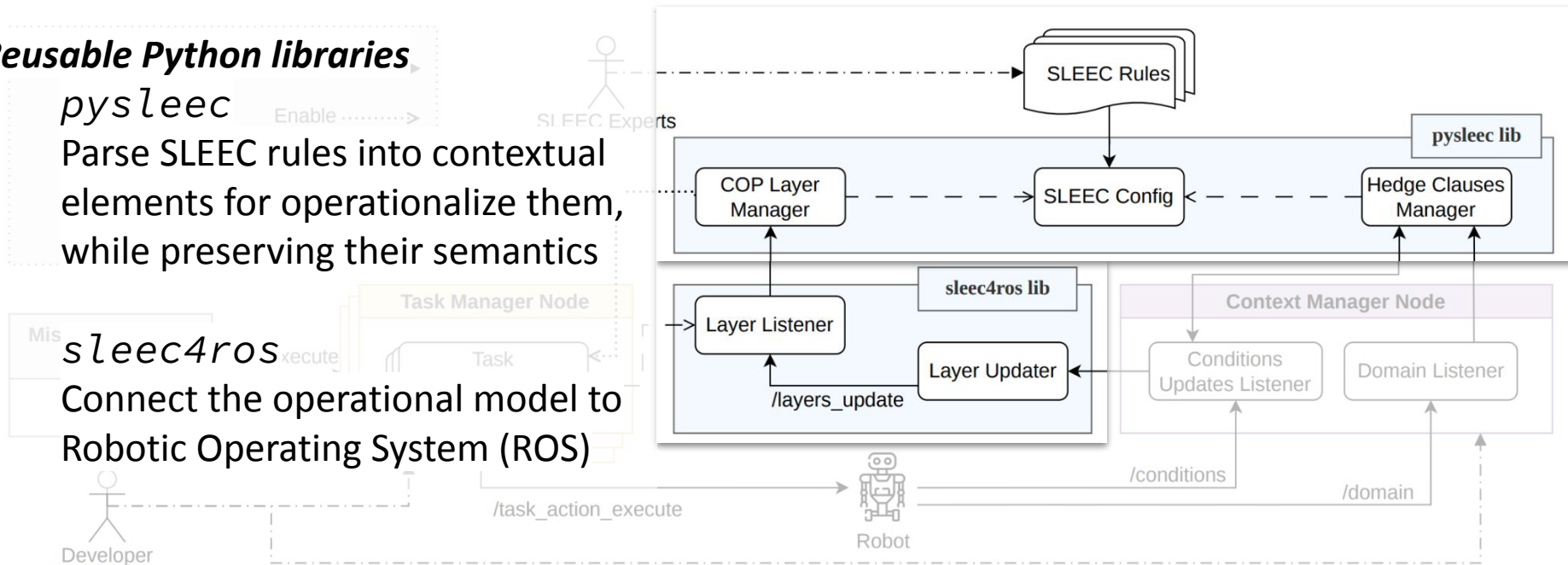
pysleec

Enable

Parse SLEEC rules into contextual elements for operationalize them, while preserving their semantics

sleec4ros

Connect the operational model to Robotic Operating System (ROS)



CO-SLEEC for ROS architecture

<https://github.com/gssi-robotics/co-sleec>



Example

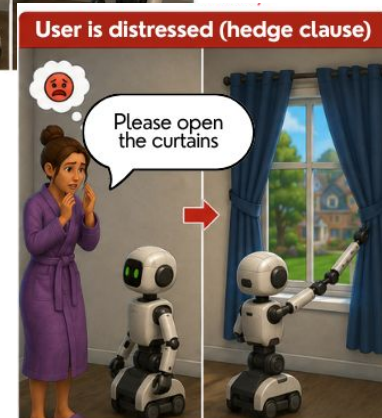
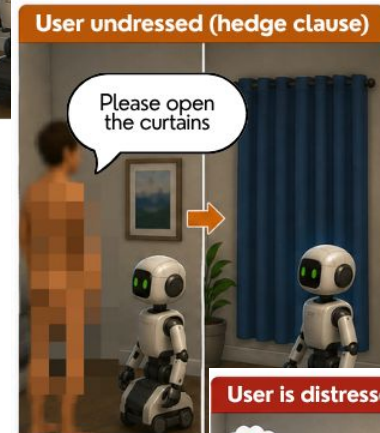
G	S
S	I

```
class OpenCurtainNode(Node):
    def __init__(self):
        super().__init__('curtain_node')
        self.create_subscription(Empty, 'open_curtains_request',
                                self.open_curtains_request, 10)

    @cop.base
    def open_curtains_request(self, msg):
        # implementation: OpenCurtains

    @cop.around(sleec_layers["dress_assist.hc1"])
    def open_curtains_request(self, msg):
        # implementation: RefuseRequest AND ExplainReason

    @cop.around(sleec_layers["dress_assist.hc2"])
    def open_curtains_request(self, msg):
        # implementation: WarnUser AND OpenCurtains
```



Empirical Evaluation

Overhead stats (data in ms)

Variant	Mean	Std. dev.	Min	Median	Max
CO-SLEEC	4.81	1.01	3.19	4.60	11.9
CS— <i>libs</i>	4.69	0.962	3.11	4.50	10.4
CS— <i>libs</i> — <i>CM</i>	4.63	0.898	3.07	4.49	10.8

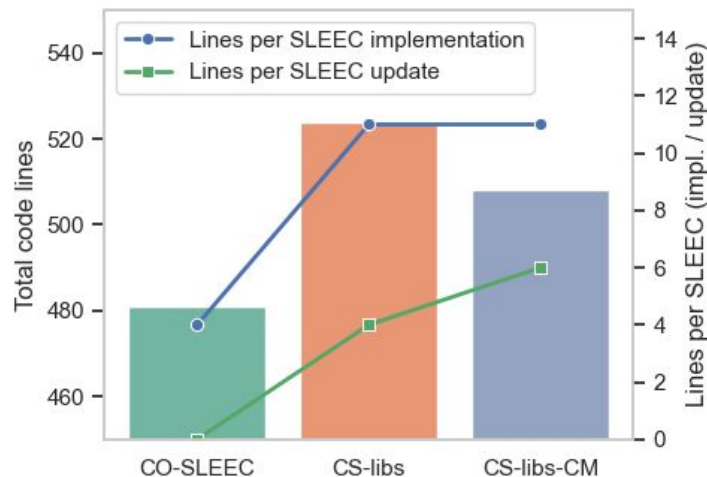
Evaluation goals:

- ❖ Correctness
- ❖ Efficiency
- ❖ Maintainability

Settings:

- ❖ 100 test cases
- ❖ 800 rule triggers
- ❖ Turtlebot 4 - Raspberry Pi 4B

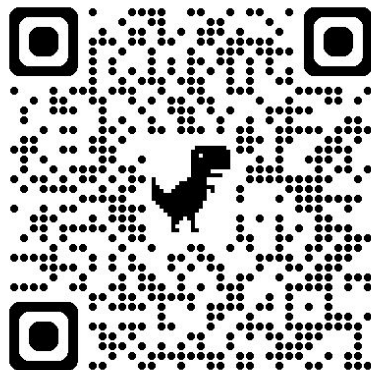
Overview of codelines across different variants



Summary



Preserve SLEEC execution semantics while making ethical behavior modular and maintainable





Questions?